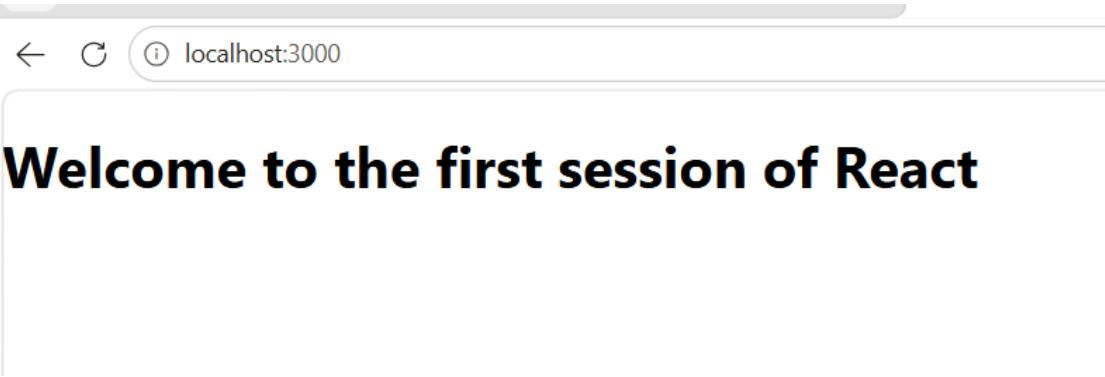


React

Exercise 1:

```
JS App.js  X
src > JS App.js > [🔗] default
1  import React from 'react';
2
3  function App() {
4    return (
5      <div>
6        <h1>Welcome to the first session of React</h1>
7      </div>
8    );
9  }
10
11  export default App;
```



Exercise 2

```
> Components > JS About.js > [🔗] default
1  import React from 'react';
2
3  class About extends React.Component {
4    render() {
5      return (
6        <div>
7          <h1>Welcome to the About page of the Student Management Portal</h1>
8        </div>
9      );
10   }
11 }
12
13 export default About;
```

```
import React from 'react';

class Contact extends React.Component {
  render() {
    return (
      <div>
        <h1>Welcome to the Contact page of the Student Management Portal</h1>
      </div>
    );
  }
}

export default Contact;
```

```
> Components > JS Home.js > [🔗] default
1  import React from 'react';
2
3  class Home extends React.Component {
4    render() {
5      return (
6        <div>
7          <h1>Welcome to the Home page of Student Management Portal</h1>
8        </div>
9      );
10   }
11 }
12
13 export default Home;
```

```
> JS App.js > App
1 import React from 'react';
2 import Home from '../Components/Home';
3 import About from '../Components/About';
4 import Contact from '../Components/Contact';
5
6 function App() {
7   return (
8     <div>
9       <Home />
10      <About />
11      <Contact />
12    </div>
13  );
14 }
15
16 export default App;
```

localhost:3000

Welcome to the Home page of Student Management Portal

Welcome to the About page of the Student Management Portal

Welcome to the Contact page of the Student Management Portal

Exercise 3

```
Components > JS CalculateScore.js > default
import React from 'react';
import '../Stylesheets/mystyle.css';

function CalculateScore(props) {
  const average = (props.total / props.goal).toFixed(2);

  return (
    <div className="score-card">
      <h1>Student Score Report</h1>
      <p><strong>Name:</strong> {props.name}</p>
      <p><strong>School:</strong> {props.school}</p>
      <p><strong>Total Marks:</strong> {props.total}</p>
      <p><strong>Number of Subjects:</strong> {props.goal}</p>
      <p className="average"><strong>Average Score:</strong> {average}</p>
    </div>
  );
}

export default CalculateScore;
```

```
> Stylesheets > # mystyle.css > .average
1 .score-card {
2   font-family: Arial, sans-serif;
3   background-color: #f0f8ff;
4   border: 2px solid #4CAF50;
5   border-radius: 10px;
6   padding: 20px;
7   width: 350px;
8   margin: 40px auto;
9   text-align: center;
10  box-shadow: 2px 2px 10px rgba(0, 0, 0, 0.15);
11 }
12
13 .score-card h1 {
14   color: #2c3e50;
15   font-size: 22px;
16   margin-bottom: 15px;
17 }
18
19 .score-card p {
20   font-size: 16px;
21   margin: 6px 0;
22   color: #333;
23 }
24
25 .average {
26   color: #4CAF50;
27   font-size: 18px;
28   margin-top: 10px;
29 }
```

```
JS App.js > App
import React from 'react';
import CalculateScore from './Components/CalculateScore';

function App() {
  return (
    <div>
      <CalculateScore
        name="Ramu"
        school="Abc school"
        total={450}
        goal={5}
      />
    </div>
  );
}

export default App;
```

Student Score Report

Name: Ramu

School: Abc school

Total Marks: 450

Number of Subjects: 5

Average Score: 90.00

Exercise 4

```
Posts.js > default
import React from 'react';

class Posts extends React.Component {
  constructor(props) {
    super(props);
    this.state = {
      posts: [],
      hasError: false
    };
  }

  loadPosts() {
    fetch('https://jsonplaceholder.typicode.com/posts')
      .then((response) => response.json())
      .then((data) => {
        this.setState({ posts: data });
      })
      .catch((error) => {
        console.error('Error fetching posts:', error);
      });
  }

  componentDidMount() {
    this.loadPosts();
  }

  componentDidCatch(error, info) {
    this.setState({ hasError: true });
    alert('An error occurred: ' + error.toString());
  }

  render() {
    if (this.state.hasError) {
      return <h2>Something went wrong while displaying the posts.</h2>;
    }

    return (
      <div>
        <h1>Blog Posts</h1>
        {this.state.posts.map((post) => (
          <div key={post.id}>
            <h2>{post.title}</h2>

```

```
> JS Post.js > default
1 class Post {
2   constructor(id, title, body) {
3     this.id = id;
4     this.title = title;
5     this.body = body;
6   }
7 }
8
9 export default Post;
```

Blog Posts

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Exercise 5

```
CohortDetails.module.css JS CohortDetails.js JS App.js X
src > JS App.js > App
1  import React from 'react';
2  import CohortDetails from './CohortDetails';
3
4  function App() {
5    return (
6      <div>
7        <h1>Cohort Details</h1>
8        <CohortDetails
9          cohortName="React Batch 1"
10         status="ongoing"
11         startDate="01-Jul-2026"
12         trainer="Anitha Kumar"
13       />
14       <CohortDetails
15         cohortName="Angular Batch 2"
16         status="completed"
17         startDate="01-Mar-2026"
18         trainer="Rahul Sharma"
19       />
20       <CohortDetails
21         cohortName="Node.js Batch 3"
22         status="ongoing"
23         startDate="15-Jun-2026"
24         trainer="Priya Menon"
25       />
26     </div>
27   );
28 }
29
30 export default App;
```

```
> # CohortDetails.module.css > dt
1  .box {
2    width: 300px;
3    display: inline-block;
4    margin: 10px;
5    padding: 10px 20px;
6    border: 1px solid black;
7    border-radius: 10px;
8  }
9
10 dt {
11   font-weight: 500;
12 }
```

```
# CohortDetails.module.css JS CohortDetails.js X JS App.js
src > JS CohortDetails.js > default
4  function CohortDetails(props) {
5    const titleColor = props.status === 'ongoing' ? 'green' : 'blue';
6
7    return (
8      <div className={styles.box}>
9        <h3 style={{ color: titleColor }}>{props.cohortName}</h3>
10       <dl>
11         <dt>Status</dt>
12         <dd>{props.status}</dd>
13         <dt>Start Date</dt>
14         <dd>{props.startDate}</dd>
15         <dt>Trainer</dt>
16         <dd>{props.trainer}</dd>
17       </dl>
18     </div>
19   );
20 }
21
22 export default CohortDetails;
```

Cohort Details

React Batch 1	Angular Batch 2	Node.js Batch 3
Status ongoing	Status completed	Status ongoing
Start Date 01-Jul-2026	Start Date 01-Mar-2026	Start Date 15-Jun-2026
Trainer Anitha Kumar	Trainer Rahul Sharma	Trainer Priya Menon

Exercise 9

```
JS ListofPlayers.js X JS IndianPlayers.js JS App.js
src > JS ListofPlayers.js > default
1 import React from 'react';
2
3 function ListofPlayers() {
4   const players = [
5     { name: "Rohit Sharma", score: 85 },
6     { name: "Virat Kohli", score: 92 },
7     { name: "KL Rahul", score: 45 },
8     { name: "Shubman Gill", score: 60 },
9     { name: "Suryakumar Yadav", score: 78 },
10    { name: "Hardik Pandya", score: 55 },
11    { name: "Ravindra Jadeja", score: 40 },
12    { name: "Rishabh Pant", score: 88 },
13    { name: "Axar Patel", score: 30 },
14    { name: "Mohammed Shami", score: 25 },
15    { name: "Jasprit Bumrah", score: 15 }
16  ];
17
18  const lowScorers = players.filter((player) => player.score < 70);
19
20  return (
21    <div>
22      <h2>List of Players (All 11)</h2>
23      <ul>
24        {players.map((player, index) => (
25          <li key={index}>
26            {player.name} - {player.score} runs
27          </li>
28        ))}
29      </ul>
30
31      <h2>Players with Score Below 70</h2>
32      <ul>
33        {lowScorers.map((player, index) => (
34          <li key={index}>
35            {player.name} - {player.score} runs
36          </li>
37        ))}
38      </ul>
39    </div>
40  );
41}
```

```
JS ListofPlayers.js JS IndianPlayers.js X JS App.js
src > JS IndianPlayers.js > IndianPlayers
1 import React from 'react';
2
3 function IndianPlayers() {
4   const teamPlayers = [
5     "Rohit Sharma", "Virat Kohli", "KL Rahul", "Shubman Gill",
6     "Suryakumar Yadav", "Hardik Pandya", "Ravindra Jadeja",
7     "Rishabh Pant", "Axar Patel", "Mohammed Shami", "Jasprit Bumrah"
8   ];
9
10  const [
11    odd1, , odd2, , odd3, , odd4, , odd5, , odd6
12  ] = teamPlayers;
13
14  const [
15    , even1, , even2, , even3, , even4, , even5
16  ] = teamPlayers;
17
18  // Merging two arrays using ES6 spread (merge feature)
19  const T20players = ["Rohit Sharma", "Virat Kohli", "Suryakumar Yadav"];
20  const RanjiTrophyPlayers = ["Sarfaraz Khan", "Yashasvi Jaiswal", "Ruturaj
21
22  const mergedPlayers = [...T20players, ...RanjiTrophyPlayers];
23
24  return (
25    <div>
26      <h2>Odd Team Players (Destructured)</h2>
27      <p>{odd1}, {odd2}, {odd3}, {odd4}, {odd5}, {odd6}</p>
28
29      <h2>Even Team Players (Destructured)</h2>
30      <p>{even1}, {even2}, {even3}, {even4}, {even5}</p>
31
32      <h2>Merged Team (T20 + Ranji Trophy)</h2>
33      <ul>
34        {mergedPlayers.map((player, index) => (
35          <li key={index}>{player}</li>
36        ))}
37      </ul>
38    </div>
39  );
40}
```

```
c > JS App.js > App
1  import React from 'react';
2  import ListofPlayers from './ListofPlayers';
3  import IndianPlayers from './IndianPlayers';
4
5  function App() {
6    const flag = true;
7
8    let content;
9    if (flag) {
10     | content = <ListofPlayers />;
11   } else {
12     | content = <IndianPlayers />;
13   }
14
15   return (
16     <div>
17       <h1>Cricket App</h1>
18       {content}
19     </div>
20   );
21 }
22
23 export default App;
```

Cricket App

List of Players (All 11)

- Rohit Sharma - 85 runs
- Virat Kohli - 92 runs
- KL Rahul - 45 runs
- Shubman Gill - 60 runs
- Suryakumar Yadav - 78 runs
- Hardik Pandya - 55 runs
- Ravindra Jadeja - 40 runs
- Rishabh Pant - 88 runs
- Axar Patel - 30 runs
- Mohammed Shami - 25 runs
- Jasprit Bumrah - 15 runs

Players with Score Below 70

- KL Rahul - 45 runs
- Shubman Gill - 60 runs
- Hardik Pandya - 55 runs
- Ravindra Jadeja - 40 runs
- Axar Patel - 30 runs
- Mohammed Shami - 25 runs
- Jasprit Bumrah - 15 runs

Exercise 10

```
JS App.js > App
import React from 'react';

function App() {
  // Single office object
  const office = {
    name: "Skyline Business Hub",
    rent: 55000,
    address: "12th Floor, MG Road, Bangalore"
  };

  // List of office objects to loop through
  const officeList = [
    { id: 1, name: "Skyline Business Hub", rent: 55000, address: "MG Road, Bangalore" },
    { id: 2, name: "Tech Park Towers", rent: 72000, address: "Whitefield, Bangalore" },
  ];

  return (
    <div>
      <h1>Office Space Rental Listings</h1>

      <h2>Featured Office</h2>
      <p><strong>Name:</strong> {office.name}</p>
      <p><strong>Address:</strong> {office.address}</p>
      <p style={{ color: office.rent < 60000 ? 'red' : 'green' }}>
        <strong>Rent:</strong> ₹{office.rent}
      </p>

      <h2>All Available Offices</h2>
      <ul>
        {officeList.map((item) => (
          <li key={item.id} style={{ marginBottom: '10px' }}>
            <strong>{item.name}</strong> – {item.address}
            <br />
            <span style={{ color: item.rent < 60000 ? 'red' : 'green' }}>
              Rent: ₹{item.rent}
            </span>
          </li>
        ))}
      </ul>
    </div>
```


Office Space Rental Listings



Featured Office

Name: Skyline Business Hub

Address: 12th Floor, MG Road, Bangalore

Rent: ₹55000

All Available Offices

- **Skyline Business Hub** — MG Road, Bangalore
Rent: ₹55000
- **Tech Park Towers** — Whitefield, Bangalore
Rent: ₹72000

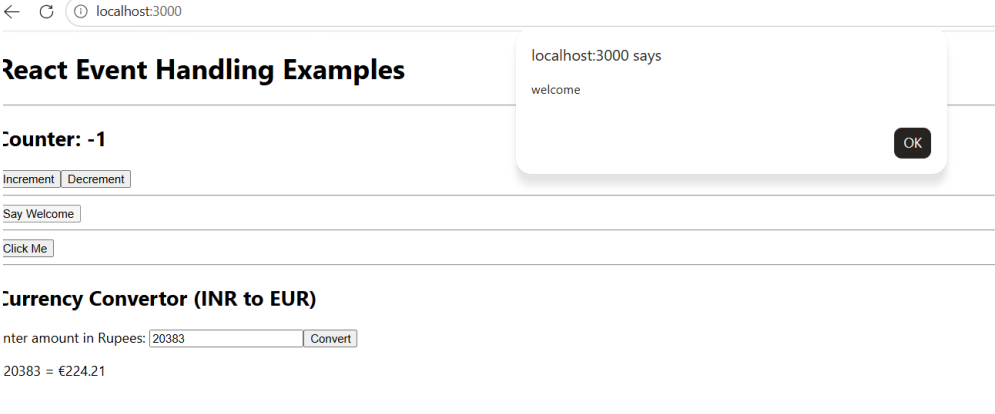
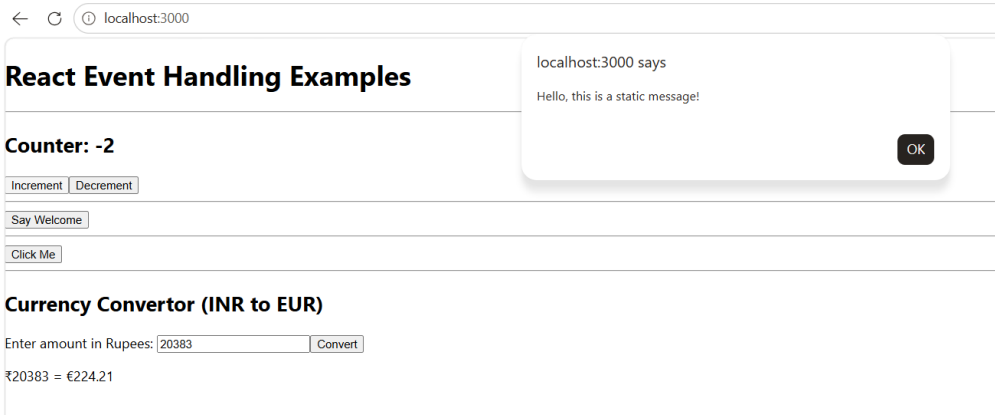
Exercise 11

```
src > JS App.js > App
1  import React from 'react';
2  import Counter from './Counter';
3  import WelcomeButton from './WelcomeButton';
4  import ClickEvent from './ClickEvent';
5  import CurrencyConvector from './CurrencyConvector';
6
7  function App() {
8    return (
9      <div>
10       <h1>React Event Handling Examples</h1>
11
12       <hr />
13       <Counter />
14
15       <hr />
16       <WelcomeButton />
17
18       <hr />
19       <ClickEvent />
20
21       <hr />
22       <CurrencyConvector />
23     </div>
24   );
25 }
26
27 export default App;
```

```
1  import React from 'react';
2
3  class WelcomeButton extends React.Component {
4    sayWelcome(message) {
5      alert(message);
6    }
7
8    render() {
9      return (
10       <div>
11         <button onClick={() => this.sayWelcome("welcome")}>
12           Say Welcome
13         </button>
14       </div>
15     );
16   }
17 }
18
19 export default WelcomeButton;
```

```
> JS ClickEvent.js > default
1 import React from 'react';
2
3 class ClickEvent extends React.Component {
4   handleClick = (event) => {
5     // 'event' here is a Synthetic Event, wrapping the native browser event
6     alert("I was clicked");
7     console.log("Synthetic event type:", event.type);
8   };
9
10  render() {
11    return (
12      <div>
13        <button onClick={this.handleClick}>Click Me</button>
14      </div>
15    );
16  }
17 }
18
19 export default ClickEvent;
```

```
src > JS App.js > App
1 import React from 'react';
2 import Counter from './Counter';
3 import WelcomeButton from './WelcomeButton';
4 import ClickEvent from './ClickEvent';
5 import CurrencyConvertor from './CurrencyConvertor';
6
7 function App() {
8   return (
9     <div>
10       <h1>React Event Handling Examples</h1>
11
12       <hr />
13       <Counter />
14
15       <hr />
16       <WelcomeButton />
17
18       <hr />
19       <ClickEvent />
20
21       <hr />
22       <CurrencyConvertor />
23     </div>
24   );
25 }
26
27 export default App;
```



Exercise 12

```
JS GuestPage.js > default
import React from 'react';

function GuestPage() {
  const flights = [
    { id: 1, name: "IndiGo 6E-204", from: "Chennai", to: "Delhi", time: "06:30 AM" },
    { id: 2, name: "Air India AI-505", from: "Mumbai", to: "Bangalore", time: "09:15 AM" },
    { id: 3, name: "SpiceJet SG-108", from: "Hyderabad", to: "Kolkata", time: "01:45 PM" }
  ];

  return (
    <div>
      <h2>Available Flights</h2>
      <p>Please log in to book a ticket.</p>
      <ul>
        {flights.map((flight) => (
          <li key={flight.id}>
            {flight.name}: {flight.from} → {flight.to} at {flight.time}
          </li>
        ))}
      </ul>
    </div>
  );
}

export default GuestPage;
```



```
src > LoginControl.js > 1
1 import React from 'react';
2 import GuestPage from './GuestPage';
3 import UserPage from './UserPage';
4
5 class LoginControl extends React.Component {
6   constructor(props) {
7     super(props);
8     this.state = { isLoggedIn: false };
9
10    this.handleLoginClick = this.handleLoginClick.bind(this);
11    this.handleLogoutClick = this.handleLogoutClick.bind(this);
12  }
13
14  handleLoginClick() {
15    this.setState({ isLoggedIn: true });
16  }
17
18  handleLogoutClick() {
19    this.setState({ isLoggedIn: false });
20  }
21
22  render() {
23    // Element variable pattern
24    let button;
25    let page;
26
27    if (this.state.isLoggedIn) {
28      button = <button onClick={this.handleLogoutClick}>Logout</button>;
29      page = <UserPage />;
30    } else {
31      button = <button onClick={this.handleLoginClick}>Login</button>;
32      page = <GuestPage />;
33    }
34
35    return (
36      <div>
37        <h1>Flight Ticket Booking</h1>
38        {button}
39      </div>
40    );
41  }
42}
```

```
function UserPage() {
  const flights = [
    { id: 1, name: "IndiGo 6E-204", from: "Chennai", to: "Delhi", time: "06:30 AM" },
    { id: 2, name: "Air India AI-505", from: "Mumbai", to: "Bangalore", time: "09:15 AM" },
    { id: 3, name: "SpiceJet SG-108", from: "Hyderabad", to: "Kolkata", time: "01:45 PM" }
  ];

  return (
    <div>
      <h2>Welcome back! Book Your Flight</h2>
      <ul>
        {flights.map((flight) => (
          <li key={flight.id}>
            {flight.name}: {flight.from} → {flight.to} at {flight.time}{' '}
            <button onClick={() => alert(`Ticket booked for ${flight.name}`)}>
              Book
            </button>
          </li>
        ))}
      </ul>
    </div>
  );
}

export default UserPage;
```

Flight Ticket Booking

Login

Available Flights

Please log in to book a ticket.

- IndiGo 6E-204: Chennai → Delhi at 06:30 AM
- Air India AI-505: Mumbai → Bangalore at 09:15 AM
- SpiceJet SG-108: Hyderabad → Kolkata at 01:45 PM

Flight Ticket Booking

Logout

Welcome back! Book Your Flight

- IndiGo 6E-204: Chennai → Delhi at 06:30 AM

Book
- Air India AI-505: Mumbai → Bangalore at 09:15 AM

Book
- SpiceJet SG-108: Hyderabad → Kolkata at 01:45 PM

Book

Exercise 13

```
JS App.js JS BookDetails.js JS BlogDetails.js JS CourseDetails.js X
src > JS CourseDetails.js > [🔍] default
1  import React from 'react';
2
3  function CourseDetails() {
4    const courses = [
5      { id: 1, name: "React Fundamentals", duration: "4 weeks" },
6      { id: 2, name: "Advanced JavaScript", duration: "6 weeks" },
7      { id: 3, name: "Node.js Basics", duration: "3 weeks" }
8    ];
9
10   return (
11     <div>
12       <h2>Course Details</h2>
13       <ul>
14         {courses.map((course) => (
15           <li key={course.id}>
16             {course.name} – {course.duration}
17           </li>
18         ))}
19       </ul>
20     </div>
21   );
22 }
23
24 export default CourseDetails;
```

```
src > JS BookDetails.js > [🔍] default
1  import React from 'react';
2
3  function BookDetails() {
4    const books = [
5      { id: 1, title: "Atomic Habits", author: "James Clear" },
6      { id: 2, title: "The Alchemist", author: "Paulo Coelho" },
7      { id: 3, title: "Deep Work", author: "Cal Newport" }
8    ];
9
10   return (
11     <div>
12       <h2>Book Details</h2>
13       <ul>
14         {books.map((book) => (
15           <li key={book.id}>
16             {book.title} by {book.author}
17           </li>
18         ))}
19       </ul>
20     </div>
21   );
22 }
23
24 export default BookDetails;
```

```
import React from 'react';

function BlogDetails() {
  const blogs = [
    { id: 1, title: "10 Tips for React Beginners" },
    { id: 2, title: "Understanding JSX" },
    { id: 3, title: "State vs Props Explained" }
  ];

  return (
    <div>
      <h2>Blog Details</h2>
      <ul>
        {blogs.map((blog) => (
          <li key={blog.id}>{blog.title}</li>
        ))}
      </ul>
    </div>
  );
}

export default BlogDetails;
```

```
1 import React from 'react';
2 import BookDetails from './BookDetails';
3 import BlogDetails from './BlogDetails';
4 import CourseDetails from './CourseDetails';
5
6 class App extends React.Component {
7   constructor(props) {
8     super(props);
9     this.state = {
10       selection: 'book',
11       showCourses: true
12     };
13   }
14
15   render() {
16
17     let bookSection;
18     if (this.state.selection === 'book') {
19       bookSection = <BookDetails />;
20     } else {
21       bookSection = null;
22     }
23
24
25     let switchResult;
26     switch (this.state.selection) {
27       case 'book':
28         switchResult = <p>You selected: Book Details view</p>;
29         break;
30       case 'blog':
31         switchResult = <p>You selected: Blog Details view</p>;
32         break;
33       case 'course':
34         switchResult = <p>You selected: Course Details view</p>;
35         break;
36       default:
37         switchResult = <p>No selection made</p>;
38     }
39
40     return (
41       <div>
42         <h1>Blogger App</h1>
43
44         <div>
45           <button onClick={() => this.setState({ selection: 'book' })}>Show Book</button>
46           <button onClick={() => this.setState({ selection: 'blog' })}>Show Blog</button>
47           <button onClick={() => this.setState({ selection: 'course' })}>Show Course</button>

```

Blogger App

Show Book

Show Blog

Show Course

Toggle Course Visibility

Book Details

- Atomic Habits by James Clear
- The Alchemist by Paulo Coelho
- Deep Work by Cal Newport

Blog section hidden

Course Details

- React Fundamentals — 4 weeks
- Advanced JavaScript — 6 weeks
- Node.js Basics — 3 weeks

You selected: Book Details view

Courses not actively selected.

Blogger App

Show BookShow BlogShow CourseToggle Course Visibility

Book Details

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Course Details

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You selected: Book Details view

Courses not actively selected.